Seminar Topics

Group 1&2 (Roll No: 1-8): Artificial Neural Networks- Deep Learning, Machine Learning Vs Deep Learning, **ANN-**Definition, Biological neural network vs artificial neural network, Architecture, Applications, Advantages and disadvantages; **The Perceptron-** Definition, Structure of the perceptron, Weight and bias, Perceptron training rule, **Multi-Layer Perceptron:** Definition, Structure of the MLP, Advantages of the MLP, Universal Approximation Theorem, Feed forward neural network, Back propagation algorithm - Weight update

Group 3 (Roll No: 9-12): Activation Functions: Linear Activation Functions: Definition, Limitations of linear activation; **Nonlinear Activation Functions-** Binary step function, Sigmoid function, Tanh function, ReLU function, Binary step vs sigmoid, advantages of ReLU in deep learning, Linear vs nonlinear activation functions

Group 4 (Roll No: 13-16): Optimization Techniques: Gradient Descent: Gradient descent algorithm (Gradient descent update rule); ; Vanishing gradient problem, **Stochastic Gradient Descent (SGD)** Steps, Advantages and Disadvantages, Gradient descent vs SGD

Group 5 (Roll No: 17-20): Regularization for Deep Learning: L1 regularization, L2 regularization, L1 vs L2 regularization, How regularization reduces overfitting; **Hyperparameter Tuning:** What is hyperparameter tuning, How hyperparameter tuning improves model performance

Group 6 & 7 (Roll No: 21-28): Convolutional Neural Networks: Definition, CNN architecture - 3 layers, Convolution operation, Need for the kernel (Example: 3×3 kernel on a 5×5 image), Pooling operation and max pooling; **Recurrent Neural Networks:** Definition, Architecture of RNN, Working of an RNN, Sequence modelling using RNNs, Batch processing in RNNs, Limitations of recurrent models, Real-world applications of RNNs; **Long Short-Term Memory (LSTM):** Definition of LSTM networks, Architecture of LSTM, Working principle of LSTM. **Generative Adversarial Networks:** Definition, Main components: generator and discriminator, Architecture of a GAN, Working of a GAN, Applications of GANs